

Excavator Simulator: sEMG-based HMI Technology Using Deep Techniques

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Abstract

This study explores low-cost wearable sEMG technology to mitigate high costs of imported VR equipment in Pakistan's VR industry. It focuses on developing a Human Machine Interface for the Metaverse, leveraging sEMG technology to enhance virtual interaction.

Introduction

This project addresses the research gap in affordable VR training solutions by developing a cost-effective sEMG-based excavator simulator. Our innovative system leverages surface electromyography (sEMG) to enhance the realism and immersion of the training experience, making it accessible and practical for widespread use. This study presents our final year's prototype, depicted in Figure 1, which demonstrates the feasibility and effectiveness of our approach in providing a high-quality training platform.

Objectives

- Designed and Manufactured Affordable Hardware:** Created and produced cost-efficient sEMG sensors and FSR-integrated hardware to ensure accessibility and affordability.
- Developed High-Performance Software:** Developed advanced software that accurately interprets muscle signals and simulates operations, ensuring a realistic training environment.
- Delivered Effective Training Results:** Provide a realistic and immersive training experience, enhancing training effectiveness, reducing costs, and improving operator skills and safety.

Flowchart

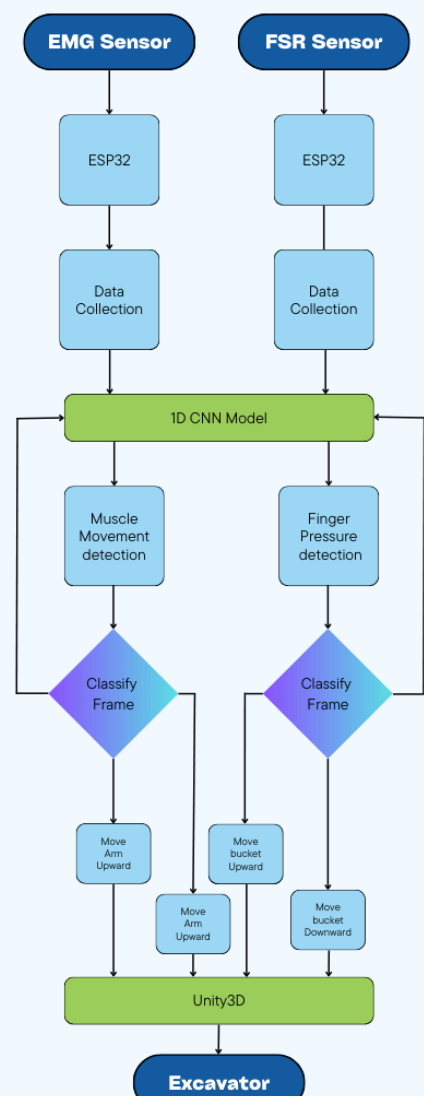


Figure 1. Flowchart showing working of the project

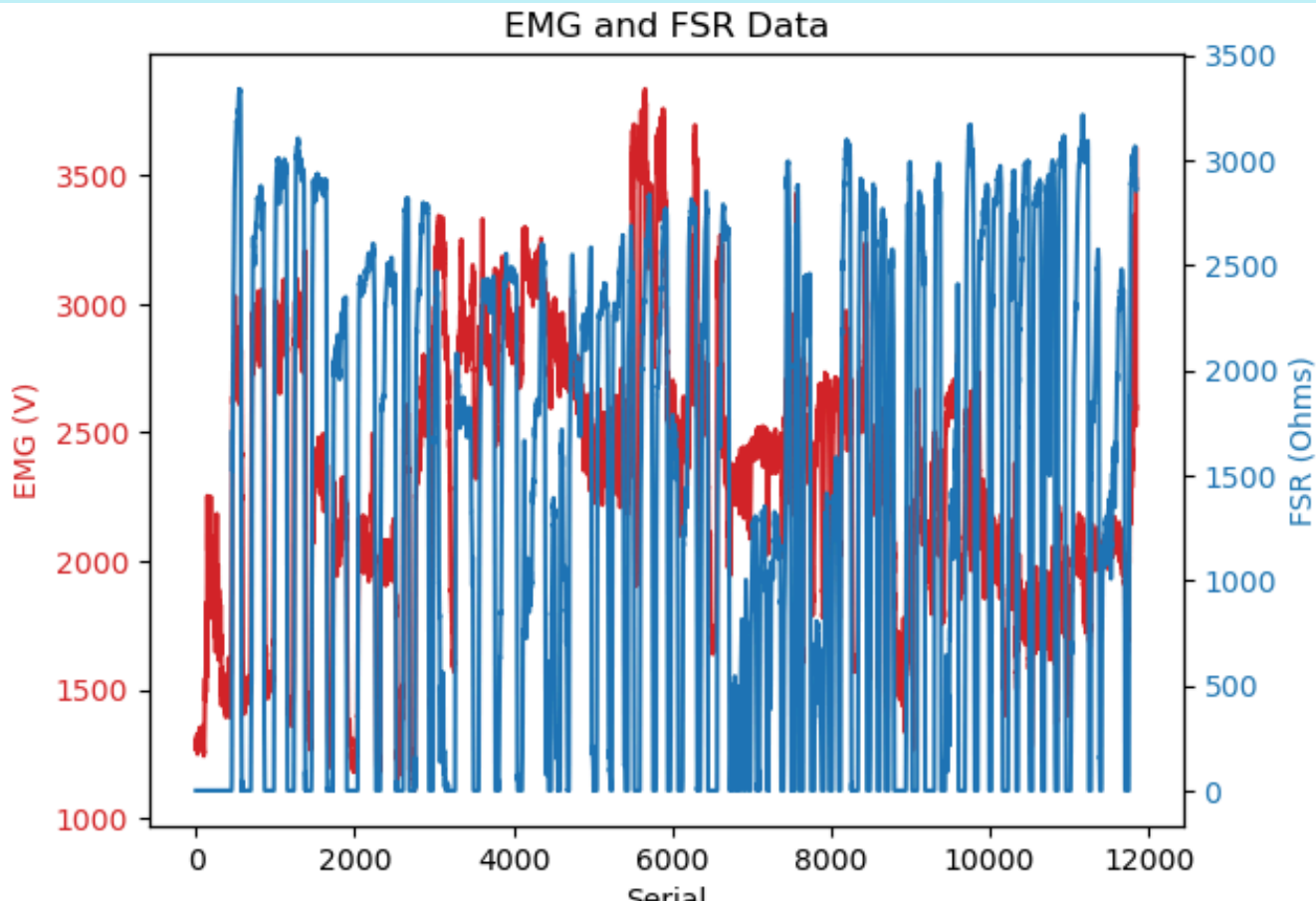


Figure 2. Waveform of EMG and FSR Sensors

Methodology

A very low-cost EMG module has been used which includes ESP32, force sensitive resistor pad and a three electrodes MS-D76 EMG sensor for the collection of the dataset.

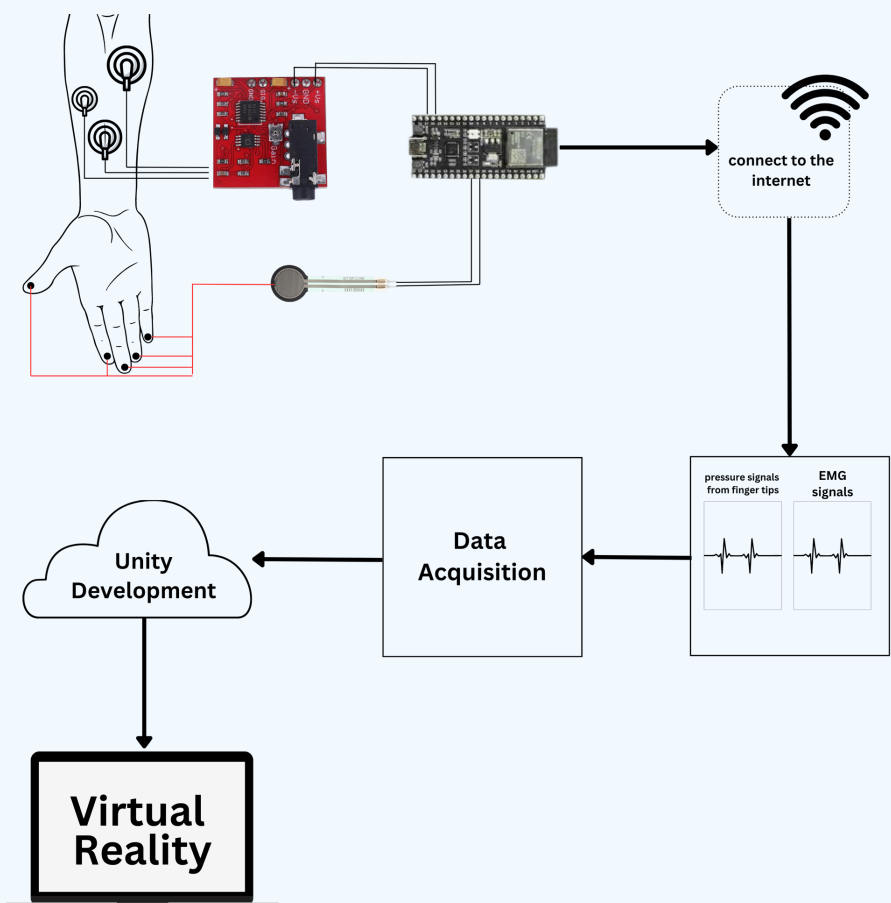


Figure 3. System Architecture

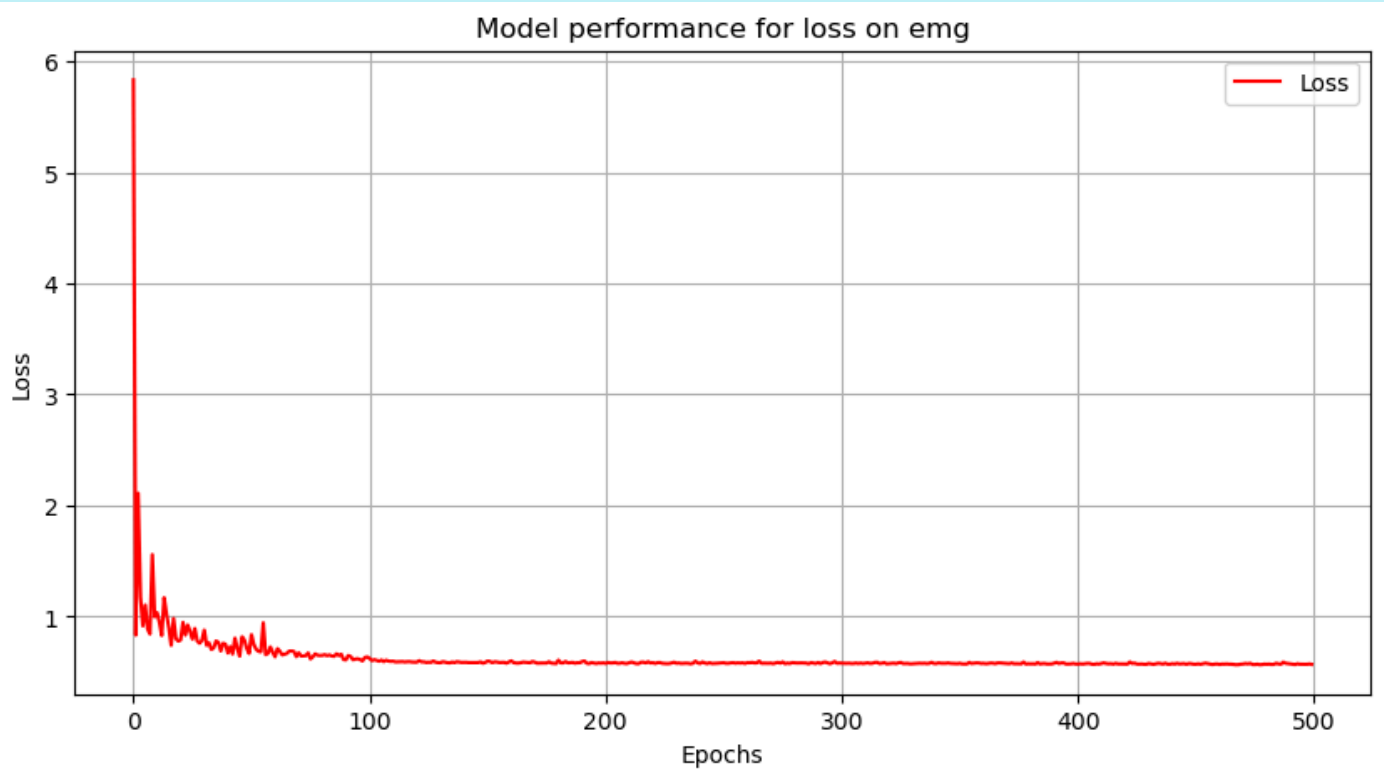


Figure 4. EMG Sensor Model Loss

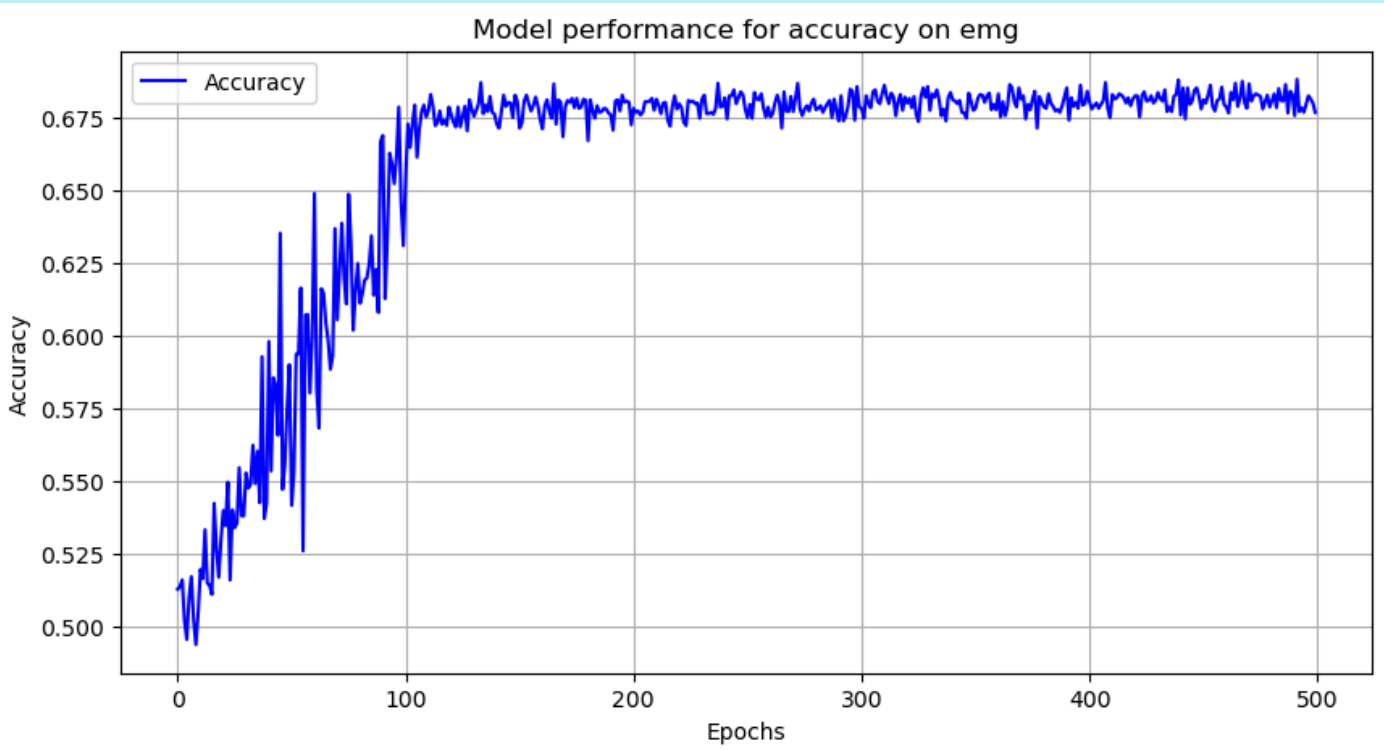


Figure 5. EMG Sensor Model Accuracy

Results

Our sensor fusion system's efficacy was evaluated by analyzing individual sensor accuracies: EMG at 71% and FSR at 98%. The combined average accuracy was calculated at 84.5%, highlighting the system's robustness and reliability in accurately capturing hand movements.

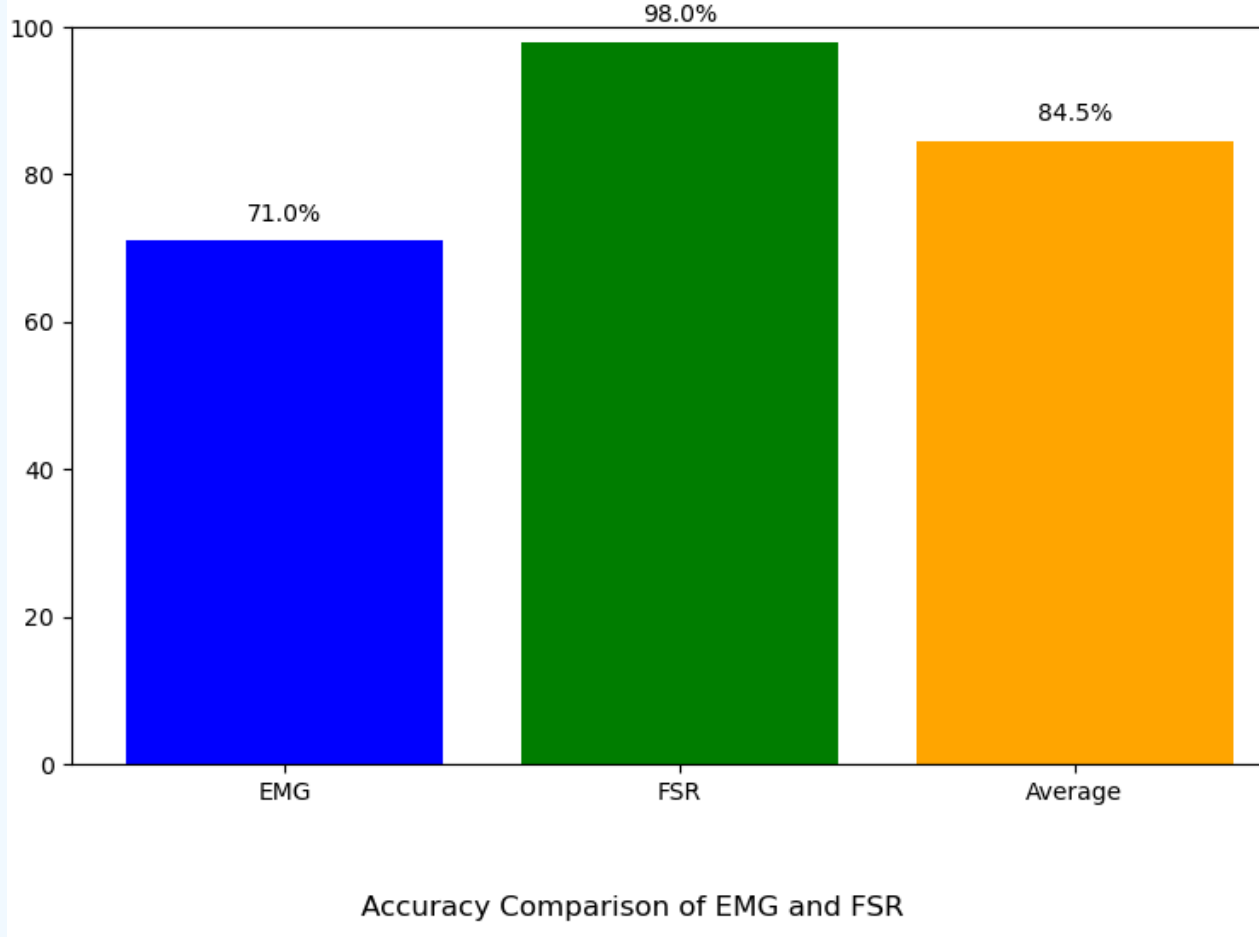


Figure 6. Waveform of EMG and FSR Sensors

Deep Learning Configurations

Altering convolutional layer architecture, such as layer count and filter size, optimizes feature extraction and pattern recognition. The figure below presents the deep learning configuration table.

No. of layers	No. of neurons	Accuracy (%)
1	32	69.01
2	64	69.46
3	128	71.02

Unity3D

The ongoing development of a Human Machine Interface (HMI) for the Metaverse incorporates surface electromyography (sEMG) technology, enhancing user interaction and control within virtual environments by detecting and interpreting muscle signals.



Figure 7. Excavator in unity

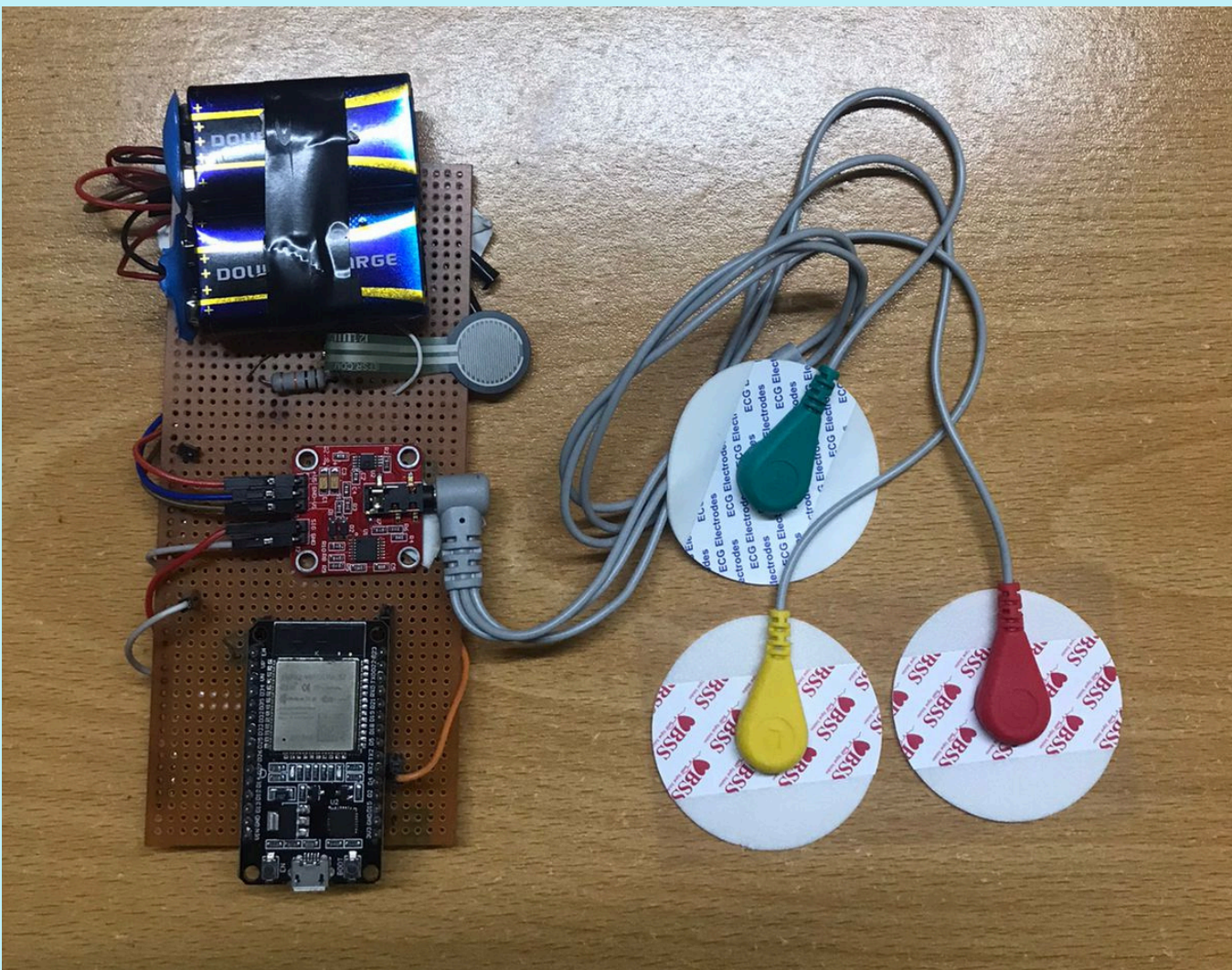


Figure 8. Final Year Project prototype

Conclusion

This study demonstrates that integrating Unity and deep learning techniques can significantly enhance the efficacy and realism of virtual training environments. The findings underscore the potential of these technologies in providing more immersive and effective training experiences, thereby contributing to advancements in virtual training methodologies.

References

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